PIYUSHA AKASH

Offensive Security Engineer · Vulnerability Researcher

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SUMMARY

Offensive Security Engineer and low-level security researcher with expertise in Windows internals, reverse engineering, exploit development, and vulnerability research. Contributor to libslirp (QEMU networking infrastructure); CVE-2026-115449 assigned by Red Hat Product Security for a NAT-isolation information-disclosure finding, with a second memory-safety finding reported and pending CVE assignment. Seeking an entry-level offensive security or penetration testing role to apply vulnerability research, systems-level tradecraft, and VAPT skills in a professional environment.

TECHNICAL SKILLS

Languages: C/C++, x86-64 Assembly, Rust, Bash Scripting, SQL

Offensive Security: Reverse Engineering, Windows Internals, AV/EDR Internals, AV/EDR Evasion, Exploit Development, Offensive Tool Engineering, Malware Analysis, Red Teaming, Penetration Testing

Security Assessment: Vulnerability Assessment, OWASP Top 10, OSINT, Social Engineering, Privilege Escalation, Post-Exploitation, Enumeration, Active Directory, MITRE ATT&CK

Security Tools: Nmap, Metasploit Framework, MSFvenom, Burp Suite, OWASP ZAP, Wireshark, Wazuh, Splunk, SIEM, Nessus, OpenVAS, FFUF, Netcat, Scapy, Social Engineering Toolkit (SET)

RE & Analysis Tools: IDA Pro, Ghidra, x64dbg, CFF Explorer, Microsoft Sysinternals Suite

Systems & Platforms: Windows, Kali Linux, Linux (Debian), VMware, Docker, Git

VULNERABILITY RESEARCH & OPEN SOURCE CONTRIBUTIONS

libslirp — freedesktop.org / QEMU User-Mode Networking Stack *C · Networking · Oct 2026*

7 patches merged to master. 2 memory-safety findings reported to Red Hat Product Security for CVE tracking. All identified via manual source code audit.

Memory-safety findings:

- Buffer underflow in tftp_handle_rrq() via empty filename — strlen underflow to SIZE_MAX causing out-of-bounds read; reported to Red Hat Product Security [Issue #100]
- NAT isolation bypass in icmp_receive() / icmp6_receive() via echo reply length desynchronization — host real IP address disclosed to guest on the raw-socket ICMP fallback path, confirmed by Red Hat as CVE-2026-115449 (CVSS 3.1: 3.5, Low) [Issue #105]

Additional hardening fixes:

- Protocol type mismatch in icmp6_receive(): IPv4 ICMP_UNREACH emitted as ICMPv6, producing malformed Time Exceeded packets [Issue #101]
- Double socket descriptor leak in fork_exec() on argument parsing failure [Issue #102]
- Socket heap memory & buffer leak in slirp_state_load() on deserialization or validation failure [Issue #103]
- Non-reentrant static variable in tcp_emu() corrupting protocol parser state across concurrent sessions [Issue #104]
- Memory leak of tftp_server_name in slirp_cleanup() [Issue #99]

gitlab.freedesktop.org/slirp/libslirp · *0x3xp.github.io/contributions/*

Control Flow Redirection via SetupAPI Callback *Independent Security Research · 2026*

- Documented methodology to subvert setupapi.dll callback routines using executable memory regions, bypassing endpoint detection telemetry monitoring conventional execution APIs
- Published C proof-of-concept alongside defensive mitigation strategies

github.com/0x3xp/SetupAPI-Callback-Redirection

PROJECTS

Artemis *C, x86-64 Assembly, Windows NT · 2026–Present*

- Low-level Windows NT offensive research framework featuring a syscall resolver supporting 14,000+ Windows builds
- Inline and indirect syscall stubs engineered for EDR telemetry bypass research

- Integrated XOR, Rolling XOR, and RC4 payload encryption schemes

github.com/0x3xp/Artemis

Native Adversary Framework *C, x86-64 Assembly · 2026–Present*

- Windows process injection research framework implementing process hollowing and thread hijacking techniques
- PEB walking and API hashing implemented entirely in x64 Assembly for stealth and portability research
- Custom PE parsers for IAT/EAT analysis enabling dynamic import resolution without Windows API

github.com/0x3xp/Native-Adversary-Framework

SENTINEX *C, Windows x64 · 2026–Present*

- Modular Windows reconnaissance framework for red team operations and adversary simulation research
- Low-noise operator shell with modular recon commands, AV/EDR exposure awareness, and LOLBins discovery
- Windows system, network, and security enumeration with minimal operational footprint

github.com/0x3xp/SENTINEX

Lethe *C, Windows Kernel (WDK), x64 · 2026–Present*

- Kernel-mode research implant using KTIMER/KDPC as execution engine; all logic runs in IO WORKITEM callbacks — no threads, no userland allocations, execution attributed to SYSTEM (PID 4)
- Research capabilities: credential access via MmCopyVirtualMemory, EPROCESS token manipulation, EDR notification callback analysis from PspCreateProcessNotifyRoutine
- Stealth research: PsLoadedModuleList unlinking, PiDDBCacheTable clearing, non-paged kernel pool operation

github.com/0x3xp/Lethe

CERTIFICATIONS

- Certified Binary Fuzzing & Reversing Professional (CBFRPro) — SecOps Group · Jul 2026
- Certified Associate Penetration Tester (CAPT) — Hackviser · Feb 2026
- Certified Windows Offensive API — Red Team Leaders · 2026
- Introduction to Linux LFS101 — The Linux Foundation · 2026

COMPETITIVE SECURITY & TRAINING

TryHackMe *Practical Cybersecurity Platform · Dec 2024–Present*

- Ranked Top 5% globally; 1st place Silver and Bronze leagues, 11th place Gold league, #1 weekly ranking in Sri Lanka
- Completed 142+ hands-on labs: VAPT, web application security, API security testing, memory forensics, OS hardening, and vulnerability chaining
- Documented vulnerabilities, attack paths, and remediation recommendations; tools: Nmap, Burp Suite, OWASP ZAP, FFUF, Netcat

PWN College *Practical Low-Level Exploitation & Reverse Engineering · Apr 2026–Present*

- Advanced exploitation and reverse engineering training in sandboxed environments
- Built custom shellcode and ROP chains; memory corruption, privilege escalation, and OS architecture

EDUCATION

Network Technician — NVQ Level 4 *Jan 2025–Present*

Technical College, Gampaha, Sri Lanka

- Information and Communication Technologies; networking fundamentals and applied network security

REFERENCES

The Principal

Technical College, Gampaha

- Institutional referee confirming course enrollment, academic compliance, and performance standards
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- Academic referee for practical technical skills, project development, and laboratory performance
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